

Pricing Theory Assignment #1

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Question 1

(a)

We need to buy stock for $\$K$ at time T . To replicate we can:

1. Borrow $\frac{K}{(1+r)^\tau}$, $\tau = T - t$
2. Buy stock, hold till forward expiry

→ This portfolio is worth

$$\Pi_T = S_T - \frac{K}{(1+r)^\tau} (1+r)^\tau = S_T - K$$

$$V_T = S_T - K \quad (\text{forward contract at time } T)$$

Both our portfolio and the forward have the same value at maturity.

∴ General expression for forward:

$$V_t = S_t - \frac{K}{(1+r)^\tau}$$

(b)

$$V_t = S_t - \frac{K}{(1+r)^\tau} = 55 - \frac{52}{(1.05)^{0.75}} = \$4.87$$

(c)

Sell forward for \$6

Time t		Time T	
Sell forward	+6	Close forward	$52 - S_T$
Buy stock	-55	Share delivered	$+S_T$
Borrow $\frac{K}{(1+r)^\tau}$	+50.13	Return loan	-52
Total	\$1.13	Total	0

Buy forward for \$4

Time t		Time T	
Buy forward	-4	Forward	$S_T - 52$
Sell stock	+55	Close short	$-S_T$
Lend	-50.13	Loan repaid	+52
Total	\$0.87	Total	0

To arb the mispriced forward we use two instruments:

1. Forward V_t
2. Replicating portfolio Π_t

We buy low & sell high, noting that the replicating portfolio always equals the no arb value of the forward. The position changes direction depending on if the market prevailing price is above (long Π) or below (short Π) the no arb price of V .

Question 2

State	0	u	d
Stock (\$)	60	75	48
Put (\$)	?	0	12

$$K = 60, \quad \text{payoff} = (60 - S_T)^+$$

(a)

Let B be the amount of bonds we have. Let N be defined similarly for the stock.

$$\begin{cases} 75N + 1.04B = 0 \\ 48N + 1.04B = 12 \end{cases}$$

which gives

$$27N = -12 \Rightarrow N = -\frac{12}{27} = -\frac{4}{9}$$

Thus

$$1.04B = -75 \left(-\frac{4}{9} \right) \Rightarrow B = 32.05$$

$$V_0 = 60 \left(-\frac{4}{9} \right) + 32.05 = \$5.38$$

(b)

$$1.04q_u + 1.04q_d = 1 \Rightarrow q_u + q_d = \frac{1}{1.04} = \frac{25}{26}$$

$$75q_u + 48q_d = 60 \Rightarrow 75q_u + 48 \left(\frac{25}{26} - q_u \right) = 60$$

$$27q_u = \frac{180}{13} \Rightarrow q_u = \frac{20}{39}, \quad q_d = \frac{35}{78}$$

(c)

$$V_0 = \sum_i V_i q_i = 0 \left(\frac{20}{39} \right) + 12 \left(\frac{35}{78} \right) = \$5.38$$

The feature via FTAP that admits no arb is the fact that the state price vector $\vec{q}$ is strictly positive.

Question 3

New derivative:

$$\delta_0 = 27, \quad \delta_u = 30, \quad \delta_d = 20$$

$$\delta_0 = 30 q_u + 20 q_d = \$24.36 \rightarrow \text{No-arb price}$$

→ With only 2 states, 2 independent assets $\Rightarrow$ 2 state prices.

Thus $\{S, B\}$ spans all possible payoffs in our market.

→ I would not purchase, as it is overpriced.